AF3625

Engineering Economics

Topic 2

Contents of Topic 2

- Consumer surplus, producer surplus and social surplus
- Price ceiling and price floor
- Price elasticity of demand
- Implications for total revenue and tax revenue

Motivations

- Why does a government regulate the price of a good or service?
- Will minimum wage increase unemployment in Hong Kong?

Economic surpluses created in the market

What is "economic surplus"? Benefit – Cost

Consumer surplus is the excess of marginal benefit (***reservation price*** or maximum price consumers are willing to pay) over marginal cost to consumers (the actual price), summed over the quantity consumed.

Producer surplus is the excess of marginal benefit to producers (the market price) over marginal cost (***reservation price*** or minimum price they are willing to accept), summed over the quantity produced.

Social surplus = Consumer surplus + Producer surplus + Tax Revenue (if tax exists, for details please refer to Topic 5)

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $20?

Market price ⇓ Reservation price ⇓

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	20	40	20	40			
2	40	60	20	20			
3	60	75	20	15			
4	80	85	20	10			
5	100	90	20	5			

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $20?

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	20	40	20	40	√	20	
2	40	60	20	20	√	0	
3	60	75	20	15	×	-5	
4	80	85	20	10	×	-10	
5	100	90	20	5	×	-15	

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $20? **CS = $20**

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	20	40	20	40	√	20	20
2	40	60	20	20	√	0	20
3	60	75	20	15	×	-5	15
4	80	85	20	10	×	-10	5
5	100	90	20	5	×	-15	-10

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $15?

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	15	40	15	40			
2	30	60	15	20			
3	45	75	15	15			
4	60	85	15	10			
5	75	90	15	5			

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $15?

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	15	40	15	40	√	25	
2	30	60	15	20	√	5	
3	45	75	15	15	√	0	
4	60	85	15	10	×	-5	
5	75	90	15	5	×	-10	

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $15? **CS = $30**

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	15	40	15	40	√	25	25
2	30	60	15	20	√	5	30
3	45	75	15	15	√	0	30
4	60	85	15	10	×	-5	25
5	75	90	15	5	×	-10	15

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $10?

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	10	40	10	40			
2	20	60	10	20			
3	30	75	10	15			
4	40	85	10	10			
5	50	90	10	5			

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $10?

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	10	40	10	40	√	30	
2	20	60	10	20	√	10	
3	30	75	10	15	√	5	
4	40	85	10	10	√	0	
5	50	90	10	5	×	-5	

Consumer Surplus (CS)

How much CS does Lisa derive from coffee per day?

When the price is $10? **CS = $45**

Q	Total cost ($)	Total benefit ($)	Marginal cost ($)	Marginal benefit ($)	Buy or not?	Increase in CS ($)	Total CS ($)
1	10	40	10	40	√	30	30
2	20	60	10	20	√	10	40
3	30	75	10	15	√	5	45
4	40	85	10	10	√	0	45
5	50	90	10	5	×	-5	40

The lower the price, the larger the CS;

The higher the price, the smaller the CS.

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $100/hr?

Market price ⇓ Reservation price ⇓

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-			
1	100	30	100	30			
2	200	100	100	70			
3	300	220	100	120			
4	400	400	100	180			
5	500	650	100	250			

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $100/hr?

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	
1	100	30	100	30	√	70	
2	200	100	100	70	√	30	
3	300	220	100	120	×	-20	
4	400	400	100	180	×	-80	
5	500	650	100	250	×	-150	

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $100/hr? **PS = $100**

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	-
1	100	30	100	30	√	70	70
2	200	100	100	70	√	30	100
3	300	220	100	120	×	-20	80
4	400	400	100	180	×	-80	0
5	500	650	100	250	×	-150	-150

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $150/hr?

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-			
1	150	30	150	30			
2	300	100	150	70			
3	450	220	150	120			
4	600	400	150	180			
5	750	650	150	250			

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $150/hr?

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	
1	150	30	150	30	√	120	
2	300	100	150	70	√	80	
3	450	220	150	120	√	30	
4	600	400	150	180	×	-30	
5	750	650	150	250	×	-150	

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $150/hr? **PS = $230**

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	-
1	150	30	150	30	√	120	120
2	300	100	150	70	√	80	200
3	450	220	150	120	√	30	230
4	600	400	150	180	×	-30	200
5	750	650	150	250	×	-100	100

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $200/hr?

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-			
1	200	30	200	30			
2	400	100	200	70			
3	600	220	200	120			
4	800	400	200	180			
5	1000	650	200	250			

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $200/hr?

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	
1	200	30	200	30	√	170	
2	400	100	200	70	√	130	
3	600	220	200	120	√	80	
4	800	400	200	180	√	20	
5	1000	650	200	250	×	-50	

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $200/hr? **PS = $400**

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	-
1	200	30	200	30	√	170	170
2	400	100	200	70	√	130	300
3	600	220	200	120	√	80	380
4	800	400	200	180	√	20	400
5	1000	650	200	250	×	-50	350

The higher the price, the larger the PS;

The lower the price, the smaller the PS.

Producer Surplus (PS)

How much PS do you derive from providing tuition service per week?

When the price is $200/hr? **PS = $400**

Hours per week	Total benefit ($)	Total cost ($)	Marginal benefit ($)	Marginal cost ($)	Sell or not?	Increase in PS ($)	Total PS ($)
0	0	0	-	-	-	-	-
1	200	30	200	30	√	170	170
2	400	100	200	70	√	130	300
3	600	220	200	120	√	80	380
4	800	400	200	180	√	20	400
5	1000	650	200	250	×	-50	350

The higher the price, the larger the PS;

The lower the price, the smaller the PS.

Maximizing Economic Surpluses (CS or PS)

Making rational decision:
Marginal approach and total approach to surplus maximization

When the price of private tuition is $200/hr, teach how many hours?

Hours per week	MB($)	MC($)	MB ≥ MC?	ΔPS ($)			
0	-	-	-	-			
1	200	30	√	170			
2	200	70	√	130			
3	200	120	√	80			
4	200	180	√	20			
5	200	250	×	-50			

Marginal approach: comparing MB with MC, you should teach only 4 hours per week. **What is the maximum PS?**

Maximizing Economic Surpluses (CS or PS)

Making rational decision:
Marginal approach and total approach to surplus maximization

When the price of private tuition is $200/hr, teach how many hours?

Hours per week	MB ($)	MC ($)	MB ≥ MC?	MB – MC ($)	TB ($)	TC ($)	TB – TC ($)
0	-	-	-	-	0	0	-
1	200	30	√	170	200	30	170
2	200	70	√	130	400	100	300
3	200	120	√	80	600	220	380
4	200	180	√	20	800	400	400
5	200	250	×	-50	1000	650	350

Total approach: looking for the largest surplus, you should teach only 4 hours per week. **The maximum PS is $400**

Maximizing Economic Surpluses (CS or PS)

Making rational decision:
Marginal approach and total approach to surplus maximization

We use marginal approach more often in our analyses.

The previous examples assume **discrete cases**, where the quantity and price are **not perfectly divisible**.

Surpluses are maximized when **MB is closest to but not less than MC** (MB≥MC).

In the next example that assumes a **continuous case**, where the quantity and price are **perfectly divisible**, ⇒ you can buy any fraction of a unit of the good by paying the corresponding amount

surpluses are maximized when **MB exactly equals MC**.

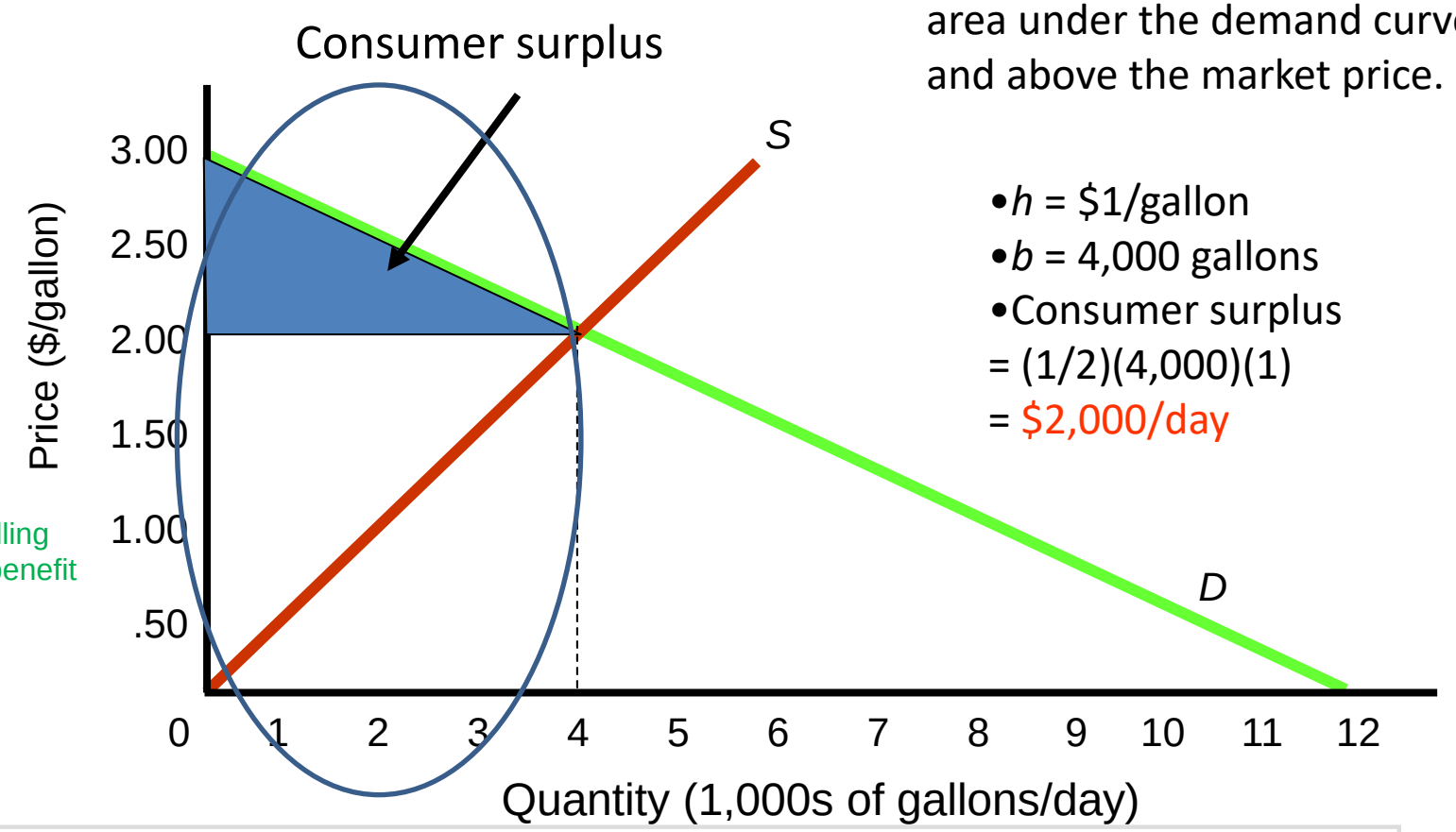
Terminologies:

"Consumers" and "buyers" can be used interchangeably.

"Producers", "suppliers" and "sellers" can be used interchangeably.

Consumer Surplus in the Market for Milk (a continuous case)

Equilibrium P = \$2 & Q = 4,000



CS generated in the market is represented by the shaded area under the demand curve and above the market price.

- h = \$1/gallon
- b = 4,000 gallons
- Consumer surplus
= (1/2)(4,000)(1)
= \$2,000/day

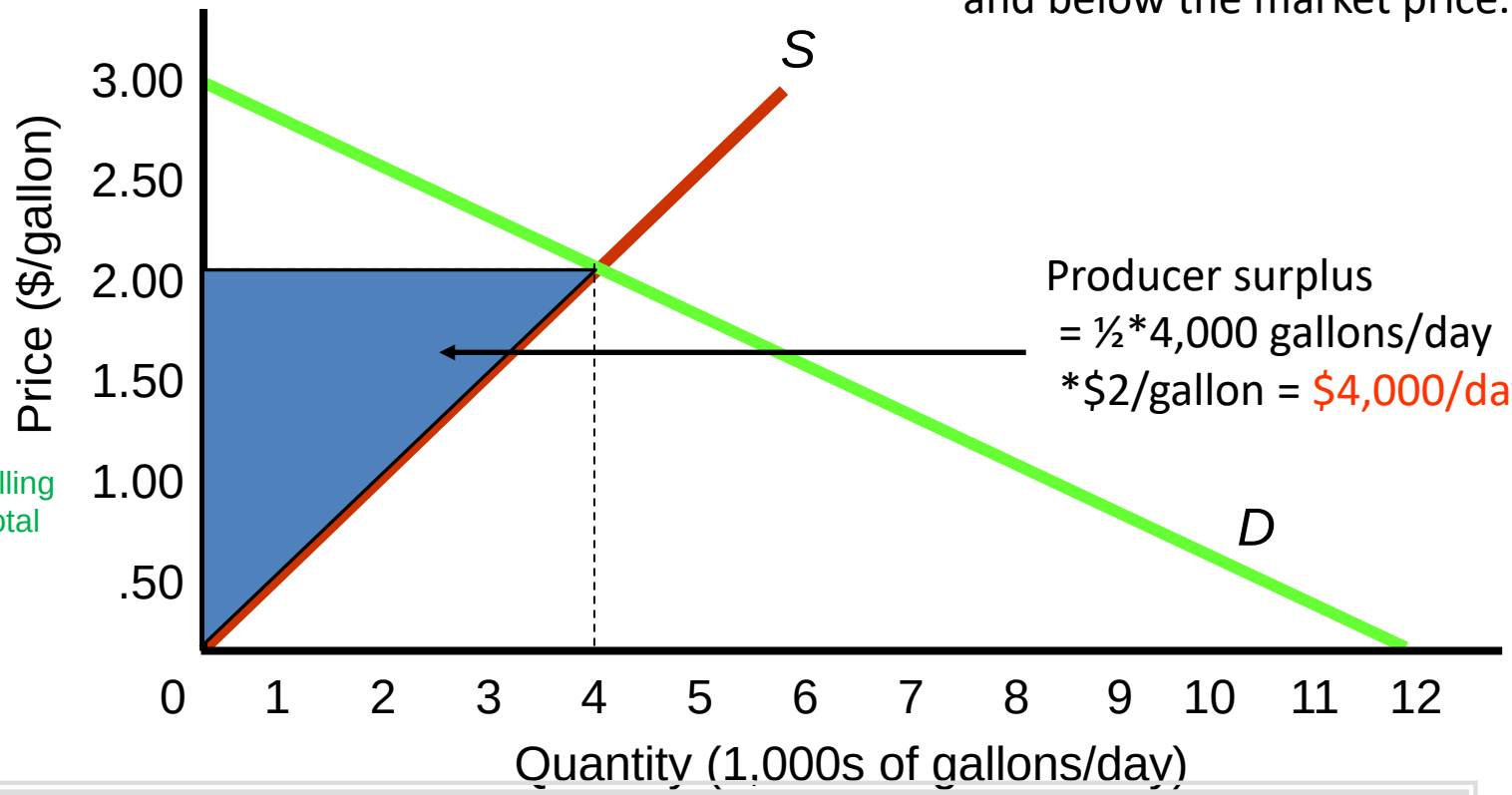
Max. amount willing to pay ⇒ Total benefit

Consumer surplus is the excess of marginal benefit (***reservation price*** or maximum price consumers are willing to pay) over marginal cost to consumers (the actual price), summed over the quantity consumed.

Producer Surplus in the Market for Milk (a continuous case)

Equilibrium P = \$2 & Q = 4,000

PS generated in the market is represented by the shaded area above the supply curve and below the market price.



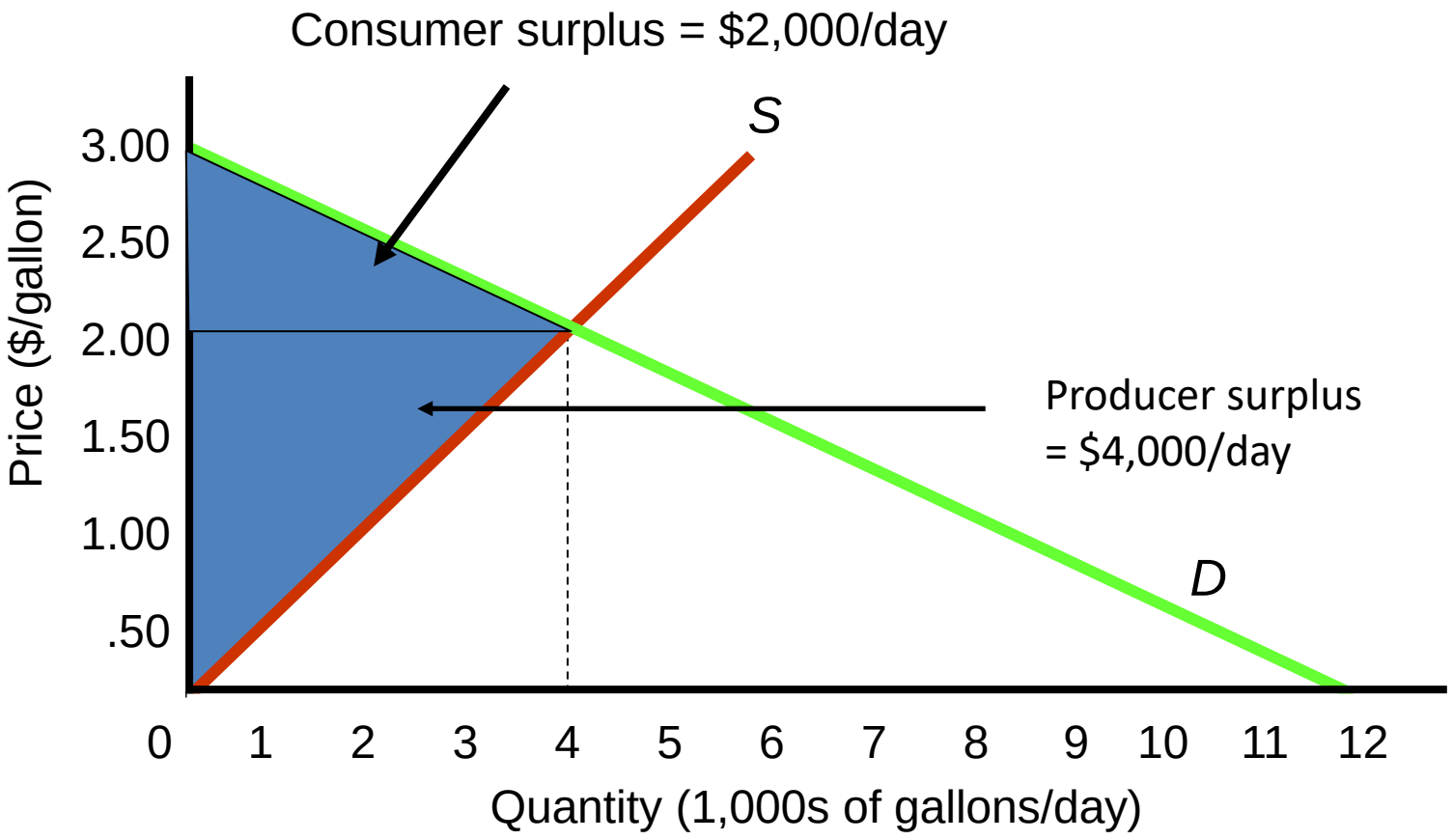
Producer surplus
= ½*4,000 gallons/day
*\$2/gallon = \$4,000/day

Min. amount willing to accept ⇒ Total cost

Producer surplus is the excess of marginal benefit to producers (the market price) over marginal cost (***reservation price*** or minimum price they are willing to accept), summed over the quantity produced.

Social Surplus in the Market for Milk (a continuous case)

Social surplus = $(2,000 + 4,000) per day = $6,000 per day



Price regulations that prevent the market from reaching equilibrium – Price ceiling and price floor

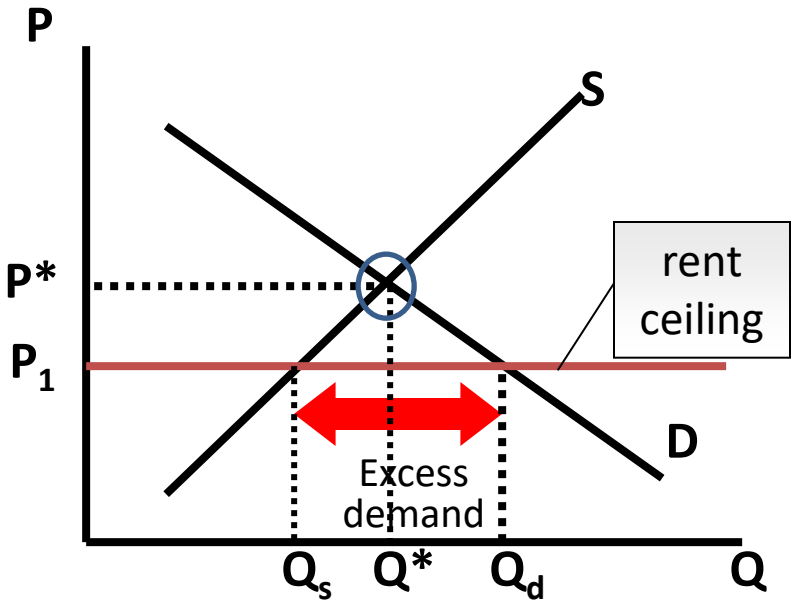
Price ceiling:

The government requires that the market price must not rise above certain level, e.g. rent control.

Why does a government regulate the price of a good or service in this case?

The equilibrium price is regarded as *too high* for the buyers.

An effective price ceiling must be set *below* the equilibrium market price. As a result, it will create an *excess demand* (or shortage) in the market.



The price ceiling is P_1 and quantity transacted in the market is Q_s (since suppliers are only willing to supply Q_s).

But at price P_1, people are willing to buy Q_d units. There will be an excess demand of $Q_d - Q_s$. $\Rightarrow$ Disequilibrium persists

Possible outcomes:

- Market is not cleared $\Rightarrow$ needs non-price rationing mechanism
- Goods sold through queuing, first come first served, rationing

Winners: Buyers who successfully buy at P_1

Losers: Buyers who are willing to pay P* but now unable to buy at P_1; Sellers (who may sell lower-quality goods in response)

Price regulations that prevent the market from reaching equilibrium – Price ceiling and price floor

Price floor:

The government requires that the market price cannot fall below certain level, e.g. minimum wage.

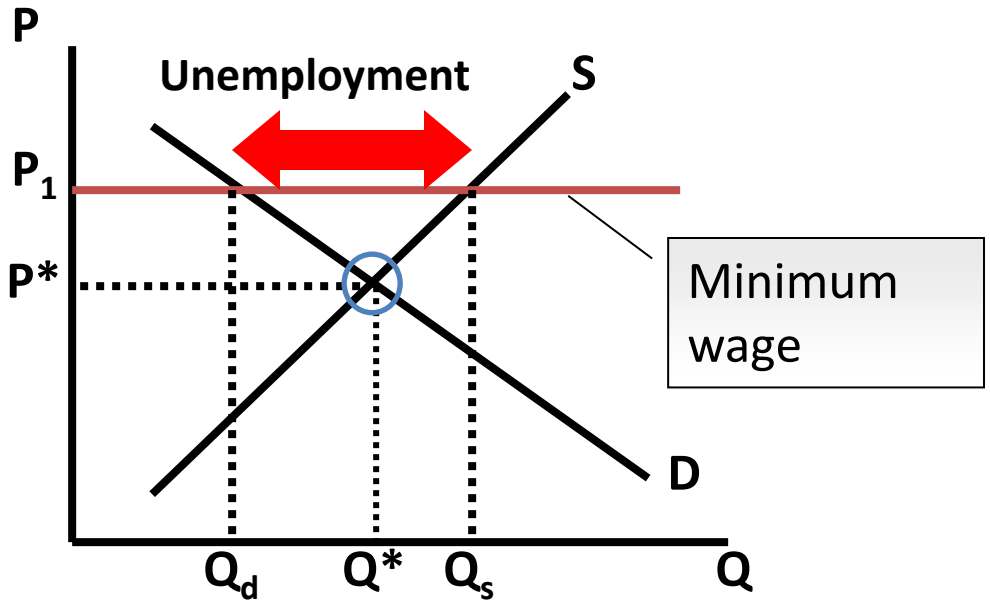
Why does a government regulate the price of a good or service in this case?

The equilibrium price is regarded as *too low* for the suppliers.

An effective price floor must be set *above* the equilibrium market price. As a result, it will create an *excess supply* (or surplus) in the market.

Critical Thinking Question:

Will minimum wage increase unemployment in Hong Kong?



The price floor is P_1 and quantity transacted in the market is Q_d (since buyers are only willing to buy Q_d units.)

But at price P_1, suppliers are willing to supply Q_s units. There will be an excess supply of $Q_s - Q_d$. **Effective *minimum* wage will increase unemployment.**

Possible outcomes: suppliers compete to have their products sold and buyers will be selective in choosing the sellers.

Winners: Sellers who successfully sell at P_1

Losers: Sellers who are willing to sell at P* but now unable to sell at P_1;
Buyers $\Rightarrow$ pay a higher price

Examples of Price Floor

Statutory Minimum Wage (SMW) has come into force in Hong Kong since 1 May 2011. With effect from 1 May 2026, the SMW rate is raised to $43.1 per hour.

Local low-skill workers

The Minimum Allowable Wage (MAW) for foreign domestic helpers (FDHs) in Hong Kong is currently set at HK$5,100 per month.



Local domestic helpers but not those from overseas.

Do you know why and how it works? You will learn more about it in Topic 5.

Critical Thinking Question: Who are the HK government trying to protect in the above two cases of price regulations?

Do Self Exercise 3 in Self-study Notes Unit 2 to analyze the overall welfare impacts of price regulations.